Midterm Group Project

Start Assignment

- Due Mar 15 by 11:59pm
- Points 30
- Submitting a file upload

Overview

For this midterm project, you will identify a policy or organizational decision problem and use a machine learning technique covered in the course to answer a set of analytic questions. Your goal is to produce insight that could realistically inform a decision by a public agency, nonprofit, or mission-driven organization, such as improving program outcomes, targeting services, allocating resources, improving compliance, or increasing operational effectiveness.

This project culminates in an 8–12 page paper and an in-class presentation of your work during a scheduled time at the start of class between March 16 and April 24.

Groups

You may work in groups of up to three people. Please do not share code or findings with anyone not in your group. There will be an anonymous peer feedback process. Members who are clearly not contributing at a professional level may receive a reduced grade.

Choosing a Topic

A strong topic typically includes:

- A concrete decision context: a policy choice, program design question, or organizational improvement goal.
- A measurable outcome: what you are trying to predict, explain, or segment (for example: uptake, completion, wait times, incidents, compliance, cost per outcome, equity gaps, retention).
- A dataset with enough rows and relevant features to support your analysis.

Good topics do not need to be flashy. The best projects answer a practical question you understand well and can make more evidence-based using machine learning tools.

I recommend choosing one of the following approaches:

- Clustering (unsupervised segmentation)
- Regression (predicting a continuous outcome)
- Classification (predicting a category or risk)

Example policy and organizational questions

Clustering

- Can we identify distinct profiles of clients in a housing assistance program to tailor outreach and supports?
- Are there clusters of neighborhoods with similar patterns of 311 requests or service gaps?

- Can we group schools or districts by enrollment trends and outcome patterns to inform resource allocation?

Regression

- What factors predict time-to-placement or length of stay in a social service program?
- What predicts program cost per participant or cost per successful outcome?
- What community factors predict public health outcomes or service utilization across neighborhoods?

Classification

- Which participants are most at risk of dropping out of a workforce training program?
- Which inspections are most likely to find violations, so we can prioritize limited inspector capacity?
- Which cases are likely to require escalation, so triage can be improved?

Deliverables

Project Proposal Due Feb 19

Submit a single-page proposal including:

1. Team member names
2. Policy or organizational decision problem and stakeholder(s) who would use the results
3. The questions you will address and the outcome(s) you will model or explore
4. Data source(s) with specifics: unit of analysis, approximate number of rows, key columns, time period, and join keys if relevant
5. Planned approach: clustering, regression, or classification, with a brief plan for evaluation
6. Any constraints (missing data, access restrictions, confidentiality) and how you will handle them

Midterm Paper Due March 15th

Write an 8–12 page paper (PDF) and submit to Canvas, along with your analysis files (notebooks/scripts) and the data source or a reproducible method to retrieve it.

Your paper should include:

- Decision context and why it matters (policy or organizational relevance)
- Data description and preparation steps
- Method explanation and why it fits the problem
- Results with clear visuals and interpretation
- Implications: what a decision-maker could do with this
- Limitations, risks, and ethical considerations (including bias and fairness where relevant)

In-class Presentation (scheduled) March 16 to April 24

All students will present their midterm project at the start of class during a scheduled time between March 16 and April 24. Your presentation should provide a concise overview of your decision context, data, method, results, and implications. More guidance on timing and format will be provided closer to the presentation dates.

Data Sources

1. **U.S. Census Bureau**: Offers comprehensive data on the U.S. population, economy, and more.
Website: https://www.census.gov (https://www.census.gov/)

2. **CIA World Factbook**: Provides information on the history, people, government, economy, and geography of countries worldwide. Website: https://www.cia.gov/the-world-factbook/ (https://www.cia.gov/the-world-factbook/)

3. **Data.gov**: The U.S. government's open data site, offering datasets across various domains.
Website: https://www.data.gov (https://www.data.gov/)

4. **Dataverse Project**: An open-source repository for research data across disciplines. Website: https://dataverse.org (https://dataverse.org/)

5. **Gapminder**: Provides global statistics with tools for data visualization. Website: https://www.gapminder.org (https://www.gapminder.org/)

6. **Google Public Data Explorer**: Allows users to explore vast datasets on various topics.
Website: https://www.google.com/publicdata/directory (https://www.google.com/publicdata/directory)

7. **HealthData.gov**: Offers datasets related to health topics from U.S. government agencies.
Website: https://healthdata.gov (https://healthdata.gov/)

8. **Kaggle Datasets**: A platform with diverse datasets for data science and machine learning projects.
Website: https://www.kaggle.com/datasets (https://www.kaggle.com/datasets)

9. **Linked Data**: A method of publishing structured data to interlink and make it more useful.
Website: https://linkeddata.org (https://linkeddata.org/)

10. **The Guardian DataBlog**: Provides datasets and analyses on various topics.
Website: https://www.theguardian.com/data (https://www.theguardian.com/data)

11. **NASA Climate Data**: Offers datasets related to climate change and environmental data.
Website: https://climate.nasa.gov (https://climate.nasa.gov/)

12. **NYC OpenData**: Provides datasets from New York City agencies. Website: https://opendata.cityofnewyork.us (https://opendata.cityofnewyork.us/)

13. **World Health Organization (WHO) Data**: Offers global health-related datasets.
Website: https://www.who.int/data (https://www.who.int/data)

14. **World Bank Open Data**: Provides free and open access to global development data.
Website: https://data.worldbank.org (https://data.worldbank.org/)

15. **UCI Machine Learning Repository**: A collection of databases for empirical machine learning research.
Website: https://archive.ics.uci.edu/ml/index.php (https://archive.ics.uci.edu/ml/index.php)

16. **DataHub.io**: A platform for discovering and sharing datasets. Website: https://datahub.io (https://datahub.io/)

17. **Maven Analytics Data Playground**: Offers free datasets for data analysis and visualization practice.
Website: https://mavenanalytics.io/data-playground (https://mavenanalytics.io/data-playground)

18. **FiveThirtyEight Datasets**: Offers datasets used in FiveThirtyEight's data-driven articles, covering topics like politics, sports, and culture. Website: **https://data.fivethirtyeight.com (https://data.fivethirtyeight.com/)**

19. **BuzzFeed News Datasets**: Provides datasets used in BuzzFeed News investigations, encompassing diverse topics. Website: **https://github.com/BuzzFeedNews/ (https://github.com/BuzzFeedNews/)**

20. **UCI Machine Learning Repository**: A collection of databases, domain theories, and datasets widely used for empirical and machine learning research. Website: **https://archive.ics.uci.edu/ml/index.php (https://archive.ics.uci.edu/ml/index.php)**

22. **Humanitarian Data Exchange (HDX)**: An open platform for sharing data across crises and organizations, managed by the United Nations Office for the Coordination of Humanitarian Affairs. Website: **https://data.humdata.org (https://data.humdata.org/)**

23. **World Bank Open Data**: Provides free and open access to global development data, including economic indicators, health statistics, and population metrics. Website: **https://data.worldbank.org (https://data.worldbank.org/)**

24. **Google Dataset Search**: A search engine specifically for datasets across the web, helping users find datasets stored across various repositories. Website: **https://datasetsearch.research.google.com (https://datasetsearch.research.google.com/)**

25. **Data USA**: A comprehensive visualization of U.S. public data, providing insights into the economy, education, and health. Website: **https://datausa.io (https://datausa.io/)**

26. **Data.gov.uk**: The UK government's open data portal, offering datasets on various topics, including government spending, crime statistics, and transportation data. Website: **https://data.gov.uk (https://data.gov.uk/)**

27. **Data Commons**: An open knowledge repository that integrates data from various sources, including the U.S. Census, World Bank, and more, providing a unified view of diverse datasets. Website: **https://datacommons.org (https://datacommons.org/)**

28. **Stanford Open Policing Project**: Standardized police stop data are available to download by location. Website: **https://openpolicing.stanford.edu/data/ (https://openpolicing.stanford.edu/data/)**

29. **FRED (Federal Reserve Economic Data) (https://fred.stlouisfed.org)**: Comprehensive U.S. economic data, including inflation, GDP, employment, and interest rates.

30. **OECD Data (https://data.oecd.org)**: Global datasets on economics, education, and public policy from the Organization for Economic Co-operation and Development.

31. **IMF Data Portal (https://data.imf.org)**: Access to international financial and macroeconomic indicators from the International Monetary Fund.

32. **NOAA Climate Data Online (https://www.ncdc.noaa.gov/cdo-web/)**: Historical climate, temperature, and weather data.

33. **USGS Earth Explorer (https://earthexplorer.usgs.gov)**: Satellite and geospatial data for land use, elevation, and environmental monitoring.

34. **EPA Open Data (https://www.epa.gov/open)**: Environmental data on air quality, water quality, and hazardous waste from the U.S. Environmental Protection Agency.

35. **Pew Research Center (https://www.pewresearch.org/download-datasets)**: Survey datasets on public opinion, social trends, and media use.

36. **Bureau of Labor Statistics (BLS)** (https://www.bls.gov/data) : Employment, wage, and price data across U.S. industries.
37. **Common Crawl** (https://commoncrawl.org) : Web-scale datasets for text and NLP research, including billions of webpages.
38. **Google Cloud Public Datasets** (https://cloud.google.com/public-datasets) : Hosted datasets for machine learning, public health, and climate science.
39. **OpenStreetMap** (https://www.openstreetmap.org) : Free and editable global geographic data.
40. **WorldPop** (https://www.worldpop.org) : Population distribution and demographic data with global coverage.
41. **OpenFlights** (https://openflights.org/data.html) : Comprehensive data on airports, airlines, and flight routes.
42. **Spotify Developer API** (https://developer.spotify.com/documentation/web-api) : Access to track, artist, and playlist data for music analysis.
43. **The Movie Database (TMDB) API** (https://developer.themoviedb.org) : Film and TV metadata for entertainment analytics.
44. **Global Forest Watch** (https://data.globalforestwatch.org) : Forestation and land use data for sustainability studies.
45. **CDC WONDER** (https://wonder.cdc.gov) : U.S. public health datasets including mortality, births, and disease prevalence.
46. **IPUMS** (https://ipums.org) : Census and survey microdata from the U.S. and around the world for demographic analysis.
47. **Esri Open Data Hub** (https://hub.arcgis.com) : Geospatial and mapping datasets from local governments and organizations.
48. **FRED Graph Data API** (https://fred.stlouisfed.org/docs/api/fred/) : API access for downloading and visualizing Federal Reserve datasets.
49. **Open Data Network** (https://www.opendatanetwork.com) : Aggregated public datasets on finance, health, and infrastructure.
50. **Global Biodiversity Information Facility (GBIF)** (https://www.gbif.org) : Worldwide biodiversity and species occurrence data.

Final Project Rubric

Criteria	Ratings						Pts
Policy and Organizational Problem Framing	**5 pts Excellent** • Clear, specific policy or organizational decision context with named stakeholder(s) and plausible use of results • Well-defined outcome(s) and analytic questions that match the decision problem • Strong rationale for why the question matters (effectiveness, efficiency, equity, compliance, risk, service quality)	**4 pts Strong** • Decision context is clear and relevant with reasonable stakeholder identification • Outcomes and questions are mostly well defined with minor gaps in specificity • Motivation is solid but could better connect to decision consequences	**3 pts Adequate** • General policy or organizational context is provided but is vague or overly broad • Outcomes and questions are stated but not tightly connected to decisions • Limited discussion of why results would change actions	**2 pts Weak** • Minimal or unclear context and stakeholders • Questions/outcomes are poorly defined or not aligned to a real decision • Weak motivation or unclear relevance	**1 pts Poor** • Problem framing is confusing or largely unrelated to policy or organizational decisions	**0 pts Missing** • No meaningful problem framing provided	5 pts
Data Source, Quality, and Preparation	**5 pts Excellent** • Data source is clearly described with unit of analysis, time period, key variables, and any joins • Thoughtful cleaning and preprocessing with justified handling of missingness, outliers, and encoding • Appropriate exploratory analysis that	**4 pts Strong** • Data is described clearly with minor missing detail • Cleaning and preprocessing are mostly appropriate with limited justification • EDA is present and useful but could be more targeted	**3 pts Adequate** • Basic description of the data and preprocessing steps • Some cleaning/EDA, but important choices are underexplained or incomplete • Limited attention to data quality concerns	**2 pts Weak** • Data description is minimal or unclear • Preprocessing is mechanical or missing key steps • Little EDA or evidence of understanding data limitations	**1 pts Poor** • Major problems in data handling or misunderstanding of dataset structure	**0 pts Missing** • No meaningful data preparation or description	5 pts

Criteria	Ratings						Pts
	reveals issues, distributions, and potential bias or measurement limits						
Method Selection and Technical Implementation	**5 pts** **Excellent** • Chooses an appropriate approach (clustering, regression, or classification) and explains why it fits the decision context • Correct implementation with sensible preprocessing for the chosen method (scaling, encoding, splits, etc.) • Demonstrates iteration or comparison across at least two reasonable model specifications or settings (for example: feature sets, hyperparameters, k values, regularization strength)	**4 pts** **Strong** • Method choice is appropriate with a reasonable explanation • Implementation is correct with minor issues or limited iteration • Some comparison across settings, but rationale could be clearer	**3 pts** **Adequate** • Method is applied correctly but with limited justification for choices • Minimal iteration or exploration of alternatives • Some technical gaps, but overall workable	**2 pts** **Weak** • Method choice is questionable for the question or poorly justified • Implementation is superficial or contains notable issues • Little evidence of iteration or careful setup	**1 pts** **Poor** • Major technical mistakes or misuse of the method	**0 pts** **Missing** • No meaningful modeling or method implementation	5 pts
Evaluation Strategy and Model Performance Evidence	**5 pts** **Excellent** • Evaluation matches the task and is clearly explained (for example: silhouette or stability for clustering;	**4 pts** **Strong** • Appropriate metrics and validation approach are used • Interpretation is mostly clear with minor gaps in linking	**3 pts** **Adequate** • Basic evaluation is present but limited in depth or clarity • Metrics are reported with minimal	**2 pts** **Weak** • Evaluation is incomplete, uses weak metrics, or is poorly explained • Validation approach is unclear or	**1 pts** **Poor** • Evaluation is misleading or largely incorrect	**0 pts** **Missing** • No meaningful evaluation of results	5 pts

Criteria	Ratings						Pts
	RMSE and residual checks for regression; confusion matrix, precision/recall, ROC-AUC for classification) • Uses an appropriate validation approach (train/test split or cross-validation where relevant) • Interprets performance in the context of decision consequences (false positives vs false negatives, thresholds, uncertainty, robustness)	to decision implications • Some robustness checks or sensitivity discussion	interpretation • Limited discussion of uncertainty, tradeoffs, or robustness	missing • Interpretation does not support conclusions			
Interpretation, Insights, and Policy or Organizational Implications	**5 pts Excellent** • Results are interpreted clearly with strong use of tables and visuals • Insights go beyond description to explain mechanisms, patterns, and relevance • Provides actionable implications tied to the stakeholder decision, including equity or distributional	**4 pts Strong** • Clear interpretation with relevant visuals • Good insights with minor gaps in depth or specificity • Implications are sensible but could be more concrete or decision-linked	**3 pts Adequate** • Basic interpretation of results • Some visuals or tables, but limited insight or explanation • Implications are general or loosely connected to decisions	**2 pts Weak** • Interpretation is shallow, repetitive, or unclear • Visuals are limited or not informative • Implications are vague or not supported by evidence	**1 pts Poor** • Misinterprets results or draws unsupported conclusions	**0 pts Missing** • No meaningful interpretation or implications	5 pts

Criteria	Ratings						Pts
	considerations when relevant						
Writing Quality, Organization, Reproducibility, and Professionalism	**5 pts** **Excellent** • Paper is well organized, readable, and professional within 8–12 pages • Figures and tables are clear, labeled, and integrated into the narrative • Code and outputs are reproducible (notebook/script runs cleanly) and methods are described sufficiently for replication • Limitations and ethical considerations are stated clearly (privacy, bias, fairness, measurement limitations)	**4 pts** **Strong** • Well written with minor issues in organization, formatting, or clarity • Visuals are mostly clear with small labeling or integration issues • Reproducibility is good with minor friction	**3 pts** **Adequate** • Understandable but uneven writing or organization • Some visuals or references are unclear or underexplained • Reproducibility is partial or requires small fixes	**2 pts** **Weak** • Significant writing, organization, or formatting problems • Visuals are confusing or poorly documented • Reproducibility issues make it hard to verify results	**1 pts** **Poor** • Major communication problems or missing key components	**0 pts** **Missing** • Paper or required materials are missing or unusable	5 pts
							Total Points: 30